

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Noel W. Clarke, Andrew J. Armstrong, Antoine Thiery-Vuillemin, et al.
Abiraterone and Olaparib for Metastatic Castration-Resistant Prostate Cancer. NEJM Evid.
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Supplementary Appendix

Supplement to Clarke NW, Armstrong A, Thiery-Vuillemin A, et al. Olaparib plus abiraterone for metastatic castration-resistant prostate cancer. N Engl J Med Evid 2022

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PROPEL INVESTIGATORS

The table below lists the principal investigator for each site who participated in the study and the number of patients randomized.

Country	Randomized by country	Participating sites	Randomized by site	Principal Investigator
Australia	62	00301	17	Angelyn Anton, Pretoria
		00305	16	Bilinski, Craig Gedye, Jeffrey Goh, Anthony Joshua, Phillip Parente, Francis Parnis, Lavinia Spain, Amanda Stevanovic, Shirley Wong
		00304	10	
		00303	7	
		00302	5	
		00306	3	
		00308	3	
		00307	1	
Belgium	5	00501	5	Nicolaas Lumen
Brazil	100	00704	21	Brigitte van Eyll, Leonardo Atem Costa, Pedro Masson Domingues, Gustavo Girotto, Juliana Janoski de Menezes, Charles A.J. Pádua, Karina Vianna
		00701	20	
		00702	18	
		00707	12	
		00703	10	
		00708	10	
		00709	9	
Canada	33	01004	11	Susan Ellard, Bryan Donnelly, Urban Emmenegger, Ricardo Fernandes, Geoffrey Gotto, Michael Kolinsky, Tamim Niazi, Fred Saad, Catherine Sperlich, Srikala Sridhar, Lori Wood
		01001	6	
		01003	4	
		01002	3	
		01008	2	
		01009	2	
		01010	2	
		01005	1	
		01006	1	
		01007	1	

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Country	Randomized by country	Participating sites	Randomized by site	Principal Investigator
Chile	44	01203	23	Pablo Gonzalez, Eugenia Loreda, Mario Orio, Anibal Salazar, Eric Orellana Ulunque
		01201	9	
		01202	7	
		01204	5	
Czech Republic	31	01902	11	Tomas Buchler, Ladislav Jarolim, Michaela Matouskova, Jiri Navratil
		01901	9	
		01903	8	
		01905	3	
France	39	02306	14	Abdel Rahmene Azzouzi, Pierre Bigot, Florence Joly, Denis Maillet, Vincent Massard, Friederike Schlürmann, Antoine Thiery-Vuillemin
		02308	9	
		02302	7	
		02301	5	
		02309	3	
		02307	1	
Germany	25	02611	7	Marinela Augustin, Susan Feyerabend, Eva Hellmis, Jörg Klier, Thomas Kretz, Stefan Machtens, Gerald Mickisch, Wolfgang Schultze-Seemann, Philipp Spiegelhalter, Friedemann Zengerling
		02601	5	
		02602	2	
		02603	2	
		02607	2	
		02608	2	
		02609	2	
		02610	2	
		02605	1	
Italy	40	04101	22	Gaetano Facchini, Franco Nolè, Sandro Pignata, Giuseppe Procopio, Mimma Rizzo, Giorgio Vittorio Scagliotti
		04102	8	
		04103	6	
		04104	3	
		04107	1	

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Country	Randomized by country	Participating sites	Randomized by site	Principal Investigator
Japan	77	04306	12	Satoshi Anai, Takayuki Goto, Yuki Yoshi Hirayama, Taro Iguchi, Satoru Kawakami, Masashi Kato, Hidefumi Kinoshita, Takeo Kosaka, Makito Miyake, Yoshiyuki Miyazawa, Atsushi Mizokami, Masayoshi Nagata, Kazuo Nishimura, Takashi Saika, Mikio Sugimoto, Hiroyoshi Suzuki, Kenichi Tabata, Naoki Terada, Hiroji Uemura, Kazuhiro Yoshimura
		04301	9	
		04302	9	
		04328	8	
		04308	7	
		04326	5	
		04317	4	
		04311	3	
		04312	3	
		04327	3	
		04332	3	
		04303	2	
		04322	2	
		04323	2	
		04324	2	
		04309	1	
		04315	1	
		04331	1	
Netherlands	28	05001	18	Niven Mehra, Robbert van Alphen, Hendrik P. van den Berg
		05003	6	
		05002	4	
Slovakia	18	06703	7	Frederico Goncalves, Ivan Mincik, Richard Polak, Roman Sokol
		06705	6	
		06702	4	
		06701	1	
South Korea	56	06001	13	Young Deuk Choi, Jun Hyuk Hong, Jae Young Joung, Cheol Kwak, Tae Gyun Kwon, Ji Youl Lee
		06003	13	
		06002	11	
		06004	10	

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Country	Randomized by country	Participating sites	Randomized by site	Principal Investigator
		06006	5	
		06005	4	
Spain	43	07001	11	Juan A. Virizuela, Daniel Castellano Gauna, Begoña Mellado González, Nuria Sala González, Maria Isabel Saez Medina, Joaquín Mateo Valderrama
		07004	11	
		07002	9	
		07003	5	
		07005	5	
		07006	2	
Turkey	72	07604	19	Ahmet Alacacioglu, Cagatay Arslan, Timucin Cil, Mustafa Ozguroglu, Mehmet Ali Nahit Sendur, Yuksel Urun
		07605	17	
		07601	11	
		07603	10	
		07606	9	
		07602	6	
UK	49	02806	15	Emma Brown, Noel Clarke, Omar Din, Carla Perna, Ahmed Shaheen
		02801	13	
		02809	8	
		02808	7	
		02805	6	
USA	74	07822	16	Andrew J Armstrong, William Aronson, James Bailen, Eric Batts, Kevin Becker, Thomas Bradley, Gary Buchschacher, Kathryn Bylow, Uzair Chaudhary, William Clark, Curtis Dunshee, Chunkit Fung, Daniel Geynisman, Pooja Ghatalia, Joelle Hamilton, Clara Hwang, Lawrence Karsh, Deepak Kilari, Eli Kirshner, Eric Knoche, Michael Lilly, Luke
		07837	9	
		07805	6	
		07815	5	
		07809	4	
		07810	3	
		07830	3	
		07836	3	
		07801	2	

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Country	Randomized by country	Participating sites	Randomized by site	Principal Investigator
		07802	2	Nordquist, Mamta Parikh, Kalpesh Patel, Christopher Pieczonka, Charles Redfern, Kristen Sanfilippo, Alton Oliver Sartor, Nasir Shahab, Neal Shore, Justin Thomas
		07804	2	
		07816	2	
		07817	2	
		07823	2	
		07824	2	
		07825	2	
		07838	2	
		07807	1	
		07811	1	
		07828	1	
		07832	1	
		07833	1	
		07835	1	
		07840	1	

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METHODS

IMAGING-BASED PROGRESSION-FREE SURVIVAL

In the protocol and statistical analysis plan for PROpel imaging-based progression-free survival is referred to as radiological progression-free survival.

METASTASIS TYPE DEFINITION

Bone only disease was defined as the presence of metastasis in the bone and no other distant site. Visceral disease was distant soft tissue metastasis in an organ (such as liver or lung) even if the patient had lesions in other metastatic sites. Other included all other patients with distant metastatic disease (e.g., patients with disease present only in distant lymph nodes).

PRESPECIFIED SUBGROUPS

Analysis of imaging-based progression-free survival was prespecified for the following subgroups:

- Metastasis (bone only, visceral, or other)
- Docetaxel treatment at metastatic hormone-sensitive prostate cancer (mHSPC) stage (yes or no)
- HRRm status subgroup (HRRm, non-HRRm, unknown) based on a ctDNA-based test (FoundationOne® Liquid CDx)
- HRRm status subgroup (HRRm, non-HRRm, unknown) based on a tissue test (FoundationOne® CDX)
- HRRm status subgroup (HRRm, non-HRRm, unknown) based on a germline blood test (Myriad My Risk™) (when data are available)
- Eastern Cooperative Oncology Group (ECOG) performance status at baseline (0 or 1)
- Age at randomization (<65 years, ≥65 years)
- Region (Asia, Europe, North and South America)
- Race (White, Black/African-American, Asian, Other)
- Baseline prostate-specific antigen (PSA) (above/below median baseline PSA of the patients across both treatment groups)

TRIAL ENDPOINTS

SECONDARY ENDPOINTS

The secondary endpoints of the trial were:

- Overall survival
- Time to first subsequent therapy or death (TFST)
- Time to pain progression
- Time to opiate use
- Time to a symptomatic skeletal-related event
- Time to second progression or death (PFS2)

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- Health-related quality of life using the Brief Pain Inventory-Short Form (BPI-SF) and Functional Assessment of Cancer Therapy (FACT) – Prostate Cancer (FACT-P) questionnaire
- Evaluation of HRR gene mutation status
- Steady-state exposure to abiraterone and its active metabolite Δ^4 -abiraterone in the presence and absence of olaparib and steady state exposure to olaparib when co-administered with abiraterone

EXPLORATORY ENDPOINTS

The exploratory endpoints of the trial were:

- Objective response rate (ORR)
- Disease control rate (DCR)
- Duration of response (DoR)
- Prostate-specific antigen (PSA) response rate
- Time to PSA progression
- To explore whether resistance mechanisms to treatment can be identified through analysis of tumor and blood samples
- Circulating tumor cell (CTC) conversion
- Longitudinal analysis of blood samples collected at regular intervals on the study (optional) in comparison to other efficacy endpoints and CTCs
- Factors that may influence development of cancer and/or response to treatment may be performed on the collected and stored archival tumor samples, blood samples (and their derivatives), and CTC samples
- Health economic impact of treatment and the disease on hospital-related resource use and health state utility

ENDPOINT DEFINITIONS

Time to first subsequent therapy or death was the time from randomization to the start of the first subsequent anticancer therapy or death from any cause, whichever was earlier. Any patient not known to have died at the time of the analysis and not known to have had a further anticancer therapy was censored at the last known time to have not received subsequent therapy. Time to first subsequent therapy or death was reported by investigator-assessment.

Time to second progression or death was the time from randomization to second progression on next-line anticancer therapy by investigator assessment of imaging-based progression, clinical symptomatic progression, or PSA progression or death.

Objective response rate (ORR) was defined as the proportion of patients with measurable disease at baseline who had a complete or partial tumor response using overall imaging-based response assessed by RECIST 1.1 (soft tissue) and PCWG-3 criteria (bone). ORR rate was reported by investigator-assessment.

Disease control rate (DCR) at 24 weeks was defined as the proportion of patients with measurable soft tissue disease at baseline who had best overall response of complete response or partial response and remained on the study without progression at 23 weeks or with a duration of stable disease of at least 23 weeks after randomization

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using overall imaging-based response assessed by RECIST 1.1 (soft tissue) and PCWG-3 criteria (bone). DCR was reported by investigator-assessment.

Duration of response (DoR) was defined as the time from tumor response (complete response or partial response) to progression or death in patients with tumor response using overall imaging-based response assessed by RECIST 1.1 (soft tissue) and PCWG-3 criteria (bone) at any time and with measurable soft tissue disease at baseline. DoR was reported by investigator-assessment.

PSA response rate was defined as the proportion of patients with a $\geq 50\%$ decrease in PSA level from baseline to the lowest post-baseline PSA result, confirmed by a consecutive PSA assessment at least 3 weeks later.

Time to PSA progression was defined as the time from randomization to PSA progression per Prostate Cancer Working Group-3 criteria.

HEALTH-RELATED QUALITY OF LIFE ASSESSMENT

As part of the measures used to assess HRQoL endpoints, the FACT-P questionnaire and the BPI-SF were administered.

The FACT-P questionnaire is a disease-specific 39-item questionnaire used to assess HRQoL and prostate cancer-specific symptoms. FACT-P Total Score ranges from 0 to 156. Change from baseline in FACT-P Total Score was analyzed using a mixed model for repeated measures and is reported in this manuscript. A change from baseline score of ≥ 10 was considered to signify an improvement in HRQoL and a change from baseline score ≤ -10 was considered to signify worsened HRQoL.^{1,2} In addition to the FACT-P total score, the FACT-G total score, trial outcome index (TOI), functional well-being (FWB), physical well-being (PWB), prostate cancer subscale (PCS), and FACT Advanced Prostate Symptom Index-6 (FAPSI-6), were also computed.

The BPI-SF is a validated, 15-item domain-specific instrument designed to assess the severity of pain and the impact/interference of pain on daily functions.³ All BPI-SF pain items including “worst pain” is scored on a 0–10 numeric rating scale with 0 indicating “no pain” and 10 indicating “worst pain imaginable”. Baseline pain score (BPI-SF Item 3 score) is reported in this manuscript.

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Figure S1. Multiplicity Strategy Maintaining Overall Type 1 Error Rate.

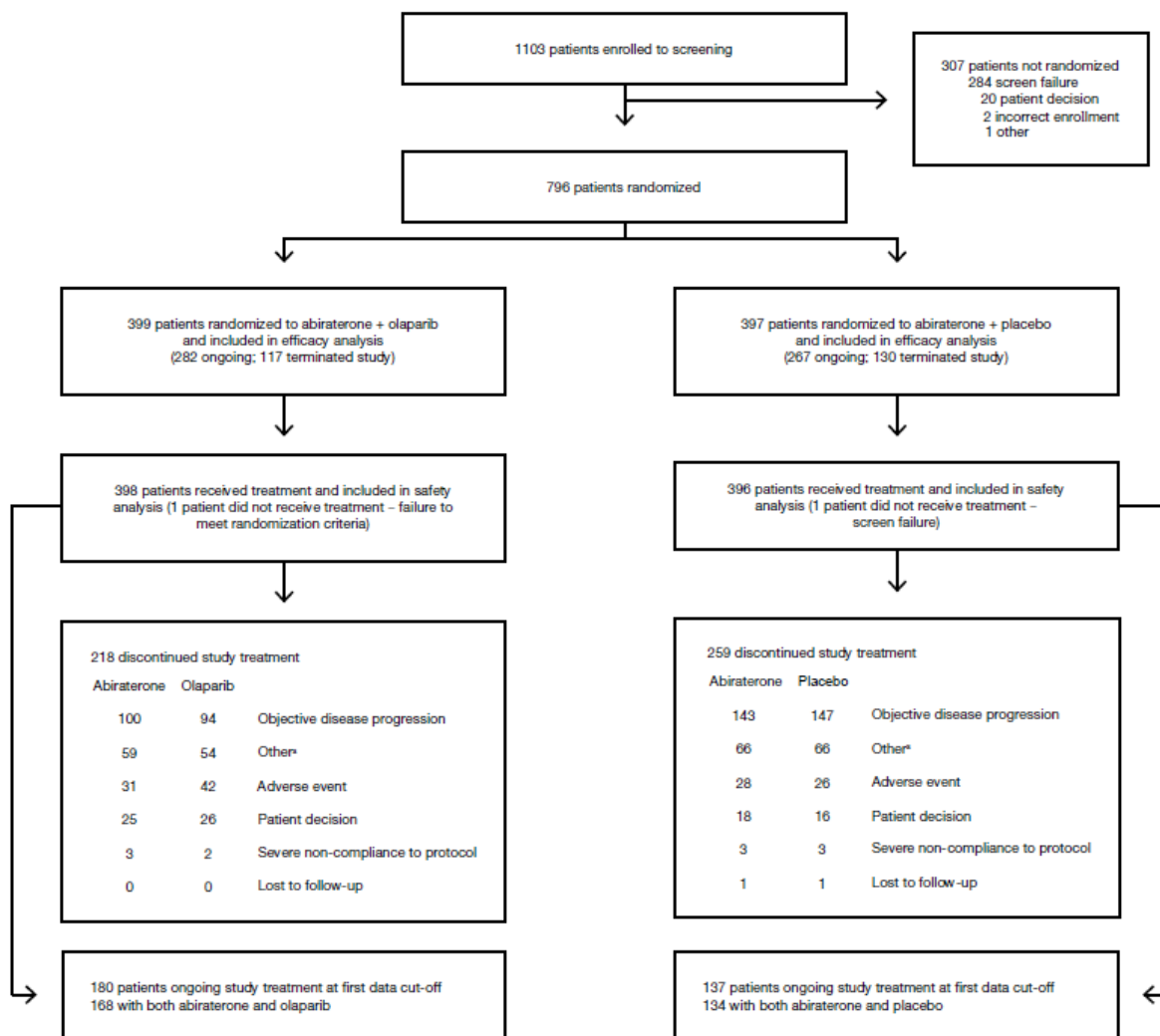


OS, overall survival; ibPFS, imaging-based progression free survival

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RESULTS

Figure S2. Patient Disposition.



PSA, prostate-specific antigen. *Other reason for discontinuation of treatment as provided by the investigator includes clinical progression, PSA progression, death, etc.

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Table S1. Characteristics of Patients at Baseline.

	Abiraterone and Olaparib (n=399)	Abiraterone and Placebo (n=397)
Median age at randomization, years (range)	69.0 (43–91)	70.0 (46–88)
Age ≥65 years at randomization, n (%)	269 (67.4)	300 (75.6)
Race, n (%)		
White	282 (70.7)	275 (69.3)
Asian	66 (16.5)	72 (18.1)
Black or African American	14 (3.5)	11 (2.8)
Other	15 (3.8)	9 (2.3)
Missing	22 (5.5)	30 (7.6)
Histology, adenocarcinoma, n (%)	398 (99.7)	397 (100)
Gleason score, n (%)		
≥8	265 (66.4)	258 (65.0)
Missing	13 (3.3)	5 (1.3)
Distant metastasis at initial diagnosis, n (%)	257 (64.4)	242 (61.0)
Missing	1 (0.3)	1 (0.3)
ECOG, n (%)		
(0) Normal activity	286 (71.7)	272 (68.5)
(1) Restricted activity	112 (28.1)	124 (31.2)
Missing	1 (0.3)	1 (0.3)
Type of progression, n (%)		
PSA progression	172 (43.1)	173 (43.6)
Imaging-based progression	92 (23.1)	73 (18.4)
Both	134 (33.6)	150 (37.8)
Missing	1 (0.3)	1 (0.3)
Prior docetaxel treatment, n (%)		
Yes	97 (24.3)	98 (24.7)
At mHSPC stage	90 (22.6)	89 (22.4)
Prior treatment with NHA*, n (%)		
Yes	1 (0.3)	0
Prior local treatment with curative intent, n (%)		
Yes	134 (33.6)	144 (36.3)
Baseline pain score (BPI-SF Item 3 score) [†] , n (%)		
0 (no pain)	133 (33.3)	137 (34.5)
>0–<4 (mild pain)	151 (37.8)	173 (43.6)
4–<6 (moderate pain)	53 (13.3)	36 (9.1)
6–10 (severe pain)	32 (8.0)	28 (7.1)
Missing	30 (7.5)	23 (5.8)
Site of disease [‡] , n (%)		
Bone	349 (87.5)	339 (85.4)
Distant lymph nodes	133 (33.3)	119 (30.0)
Locoregional lymph nodes	82 (20.6)	89 (22.4)
Prostate and adjacent structures	47 (11.8)	46 (11.6)

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	Abiraterone and Olaparib (n=399)	Abiraterone and Placebo (n=397)
Respiratory (including lung)	40 (10.0)	42 (10.6)
Liver	15 (3.8)	18 (4.5)
HRRm status [§]		
Based on ctDNA, n (%)		
HRRm	98 (24.6)	100 (25.2)
Non-HRRm	269 (67.4)	267 (67.3)
HRRm unknown	32 (8.0)	30 (7.6)
Based on tumor tissue test, n (%)		
HRRm	62 (15.5)	56 (14.1)
Non-HRRm	207 (51.9)	210 (52.9)
HRRm unknown	130 (32.6)	131 (33.0)
Based on aggregate, n (%)		
HRRm	111 (27.8)	115 (29.0)
Non-HRRm	279 (69.9)	273 (68.8)
HRRm unknown	9 (2.3)	9 (2.3)
BRCAm prevalence		
Based on aggregate, n (%)		
BRCA1	9 (2.3)	3 (0.8)
BRCA2	38 (9.5)	35 (8.8)
Median time from initial diagnosis to randomization, months (range)	33.6 (4–288)	39.5 (1–279)
Baseline S-prostate specific antigen, median (µg/L), (IQR)	17.90 (6.09–67.00)	16.81 (6.26–53.30)
S/P-alkaline phosphatase, median µkat/L (IQR)	1.88 (1.35–3.38)	1.83 (1.30–3.35)
Baseline S/P-lactate dehydrogenase, median µkat/L (IQR)	3.63 (3.07–4.69)	3.50 (2.93–4.42)
Baseline hemoglobin, median g/L, (IQR)	132.0 (123.0–139.0)	133.0 (124.0–140.0)
Baseline albumin, median g/L (IQR)	43.0 (40.0–45.0)	42.0 (39.0–45.0)
Baseline creatinine, median µmol/L (IQR)	78.0 (69.0–88.4)	77.8 (69.0–88.4)

Data derived from electronic case report forms. *Treatment with next-generation hormonal agents (except abiraterone) was allowed, provided patients had not had PSA, clinical or imaging-based progression during the treatment and the treatment was stopped at least 12 months before randomization. †Baseline pain score is based on a patient completing the BPI-SF questionnaire item 3 (worst pain) at least once during the 7 day baseline period and is presented as an average. ‡Investigators could select more than one site of disease. Entries for “Other locally advanced sites”, “Other distant sites” and “Other” have been excluded. §HRRm: Any deleterious or suspected deleterious HRR gene mutation detected; non-HRRm: no deleterious or suspected deleterious HRR gene mutation detected; HRRm unknown: patients for whom mutation testing was not performed or where mutation testing failed due to insufficient quantity or quality of sample, or technical failure at sequencing or post-sequencing steps on analysis

BPI-SF, Brief Pain Inventory-Short Form; ECOG, Eastern Cooperative Oncology Group; IQR, interquartile range; NHA, next-generation hormonal agent; PSA, prostate specific antigen.

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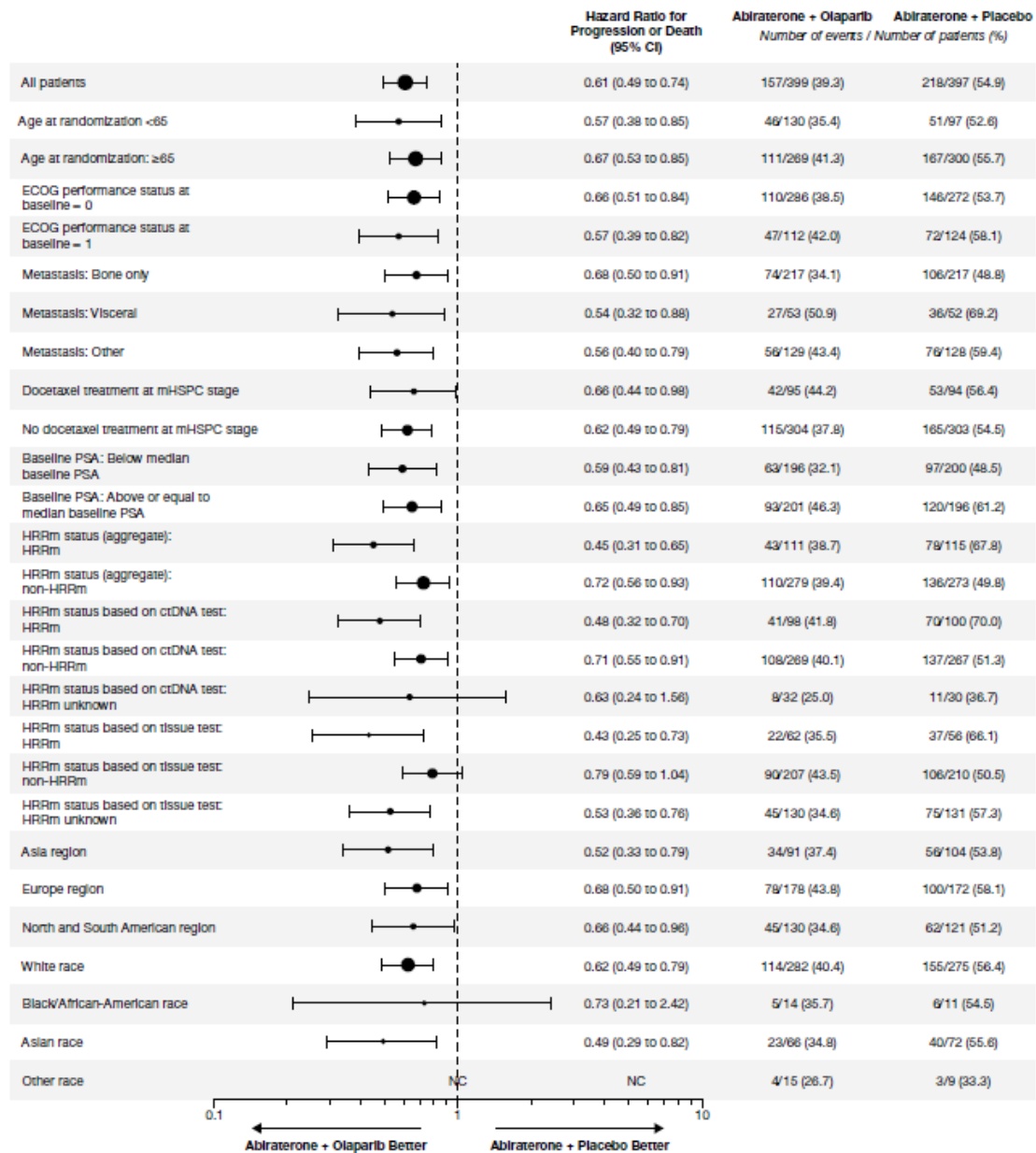
REPRESENTATION

Table S2. Representativeness of Study Participants

Disease under consideration	Prostate cancer
Special considerations related to:	
Sex and gender	Anyone with a prostate can develop prostate cancer
Age	The risk of prostate cancer increases with age
Race or ethnic group	Black people in the USA and the Caribbean have the highest incidence rates of prostate cancer globally
Geography	The highest incidence rates of prostate cancer are in Northern and Western Europe, the Caribbean, Australia/New Zealand, Northern America, and Southern Africa. The lowest incidence rates of prostate cancer are in Asia and Northern Africa.
Other considerations	Not applicable
Overall representativeness of this trial	Only patients born with a prostate could be recruited into PROpel, therefore, we consider this trial to be representative with regards to sex and gender. Patients were between 43 and 91 years old, with a median age of 69 years, which is consistent with increased risk of prostate cancer as people age. In Brazil and the USA, the participating countries where one might expect the highest representation of Black prostate cancer patients, ~10% of Brazilian patients and 15% USA patients are Black. In PROpel there were a total of 100 patients from Brazil and 74 from the USA. Based on the rates described in literature, one would expect 21 Black patients to be recruited into our trial: 10 Brazilian patients and 11 patients from the USA. We recruited 18 Black patients from Brazil, five Black patients from the USA, one Black patient from Canada and one Black patient from Spain. This was a multicenter trial across North America, South America, Europe, Asia and Oceania. No patients were enrolled in Africa.

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Figure S3. Forest Plot of Subgroup Analysis of Imaging-based Progression-Free Survival (Blinded Independent Central Review).



Data derived from Interactive Voice/Web Response System stratification variables. Analysis included stratification factors selected in the primary pooling strategy as covariates. Each subgroup analysis was performed using a Cox proportional hazards model that contains a term for treatment, factor and treatment by factor interaction. The size of a circle is proportional to the number of events. Subgroup categories with fewer than 5 events in either treatment group have NC presented. In the PSA subgroups patients with no baseline assessment were excluded from the analysis.

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Table S3. Therapies Post-Discontinuation of Study Treatment.

	Abiraterone and Olaparib (N=132)	Abiraterone and Placebo (N=173)
Cytotoxic chemotherapy	91 (68.9)	130 (75.1)
Hormonal therapy	46 (34.8)	56 (32.4)
Immunotherapy	11 (8.3)	14 (8.1)
Targeted therapy	6 (4.5)	14 (8.1)
Systemic therapy	1 (0.8)	5 (2.9)
PARP inhibitor	0	3 (1.7)
Other	3 (2.3)	7 (4.0)

Data are n (%). Patients could be counted in more than one anticancer therapy

Table S4. Objective Response.

	Abiraterone and Olaparib (n=161)	Abiraterone and Placebo (n=160)
Objective response* – n (%)	94 (58.4)	77 (48.1)
Complete response	7 (4.3)	10 (6.3)
Partial response	87 (54.0)	67 (41.9)
Stable disease ≥11 weeks, n (%)	42 (26.1)	45 (28.1)
Progressive disease, n (%)	22 (13.7)	31 (19.4)
Non-evaluable, n (%)	3 (1.9)	7 (4.4)
Disease control rate, [†] n (%)	132 (82.0)	121 (75.6)
Median duration of response, months	18.2	12.9
(95% CI)	(12.81, NC)	(10.28, 18.46)
Median time to objective response, months (95% CI)	2.2 (1.91, 3.65)	2.0 (1.94, 3.68)

*Imaging-based objective response rate assessed based on investigator-assessed RECIST and bone scan data (using all scans regardless of whether they were scheduled or not) in patients with measurable disease. Response did not require confirmation. [†]Disease control rate at 24 weeks was defined as the best overall response of complete response or partial response and remaining on study without progression at 23 weeks or having a duration of stable disease of at least 23 weeks after randomization (24 weeks minus 1 week to allow for an early assessment within assessment window).

CI, confidence interval; NC, not calculable; RECIST, Response Evaluation Criteria in Solid Tumors.

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Table S5. Efficacy Summary.

	Abiraterone and Olaparib (n=399)	Abiraterone and Placebo (n=397)	HR (95% CI)
Primary endpoint			
Imaging-based progression-free survival by investigator assessment			
Median, months	24.8	16.6	0.66 (0.54–0.81)
Sensitivity analysis of primary endpoint			
Imaging-based progression-free survival by blinded independent central review			
Median, months	27.6	16.4	0.61(0.49–0.74)
Key secondary endpoints			
Overall survival			
Median, months	NR	NR	0.86 (0.66–1.12)
Other secondary endpoints			
Time to first subsequent therapy or death			
Median, months	25.0	19.9	0.74 (0.61–0.90)
Time to second progression on next-line anticancer therapy			
Median, months	NR	NR	0.69 (0.51–0.94)
Exploratory endpoints			
ORR, n (%)	94 (58.4)	77 (48.1)	
Confirmed PSA response, n (%)	315 (79.3)	274 (69.2)	
Time to PSA progression, months	NR	12.0	0.55 (0.45–0.68)

CI, confidence interval; HR, hazard ratio; NR, not reached; ORR, objective response rate; PSA, prostate specific antigen.

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SAFETY

Table S6. Adverse Events of Cardiac Failure SMQ.

	Abiraterone and olaparib (N=398)	Abiraterone and placebo (N=396)
Cardiac failure events, n (%)	6 (1.5)	5 (1.3)
Acute pulmonary oedema	0	1 (0.3)
Cardiac failure	2 (0.5)	2 (0.5)
Cardiac failure acute	1 (0.3)	0
Cardiac failure chronic	0	1 (0.3)
Cardiac failure congestive	1 (0.3)	1 (0.3)
Ejection fraction decreased	1 (0.3)	0
Left ventricular failure	2 (0.5)	0
Pulmonary edema	0	2 (0.5)

Patients were counted once for each type of adverse event.

Table S7. Embolic and Thrombotic Adverse Events, Arterial SMQ.

	Abiraterone and olaparib (N=398)	Abiraterone and placebo (N=396)
Embolic and thrombotic events, n (%)	8 (2.0)	10 (2.5)
Acute myocardial infarction	3 (0.8)	3 (0.8)
Ischemic stroke	1 (0.3)	4 (1.0)
Myocardial infarction	1 (0.3)	1 (0.3)
Peripheral arterial occlusive disease	2 (0.5)	0
Peripheral embolism	0	1 (0.3)
Transient ischemic attack	1 (0.3)	1 (0.3)

Patients were counted once for each type of adverse event.

These data are strictly confidential and should not be shared with anyone unauthorized to view them

Table S8. Adverse events of pulmonary embolism

	Abiraterone and olaparib (N=398)	Abiraterone and placebo (N=396)
All pulmonary embolism events, n %	26 (6.5)	7 (1.8)
Exposure adjusted (1000 patient years)*	49.1	13.8
Grade ≥3	26 (6.5)	7 (1.8)
SAEs	13 (3.3)	3 (0.8)
Causally related AE (olaparib/placebo)	7 (1.8)	2 (0.5)
Interruption of olaparib/placebo because of an adverse event	8 (2.0)	1 (0.3)
Interruption of abiraterone because of an adverse event	6 (1.5)	1 (0.3)
Dose reduction of olaparib/placebo due to an adverse event	0	0
Dose reduction of abiraterone due to and adverse event	0	0
Discontinuation of olaparib/placebo due to an adverse event	0	1 (0.3)
Discontinuation of abiraterone due to an adverse event	0	1 (0.3)
Death due to an adverse event	1 (0.3)	0

Patients were counted once for each type of adverse event. Adverse events with an onset date, or worsening, on or after the date of first dose and up to and including 30 days following discontinuation of randomized treatment were included.*Number of patients with that AE divided by the sum of the duration of therapy (for patients without the AE) and the time to the AE (for patients with the AE) in each group multiplied by 1000.

PULMONARY EMBOLISM ADVERSE EVENTS

There is currently no evidence to support a potential biologic-pharmacologic plausibility between olaparib and venous thromboembolic events. The incidence of venous thromboembolic events observed in PROpel is consistent with the incidence reported in prostate cancer patients from literature. The majority of patients recovered from the event and were able to continue olaparib treatment based on standard medical practice.

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SUMMARY OF BONE MARROW OR BLOOD CYTOGENETIC ANALYSIS

Investigators were asked to consider bone marrow and/or blood cytogenetic analysis if any blood parameters remained clinically abnormal after 4 weeks of interruption of 'olaparib/placebo', according to standard hematological practice and as clinically indicated, as per the following in the study protocol:

- Section 8.2.1.2 (Bone marrow or blood cytogenetic)
- Section 8.4.4.1 (Management of hematological toxicity)
- Table 1 (Schedule of assessments – Global study)

Within PROpel, there was one record of a marrow biopsy being performed as part of a "Myelodysplastic syndromes check-up" (text as entered by the site), for which the result was "No pathological findings".

DISCUSSION

ctDNA TESTING

ctDNA testing offers many advantages as a diagnostic tool compared to the current gold standard tumor tissue test but may potentially report false-positive and false-negative cases. Of the 552 patients in the non-HRRm aggregate subgroup, 186 patients had a ctDNA result only and did not have a tumor tissue test result. Studies to date indicate there is a low percentage of HRR gene mutations, for example mutations in *BRCA1* and *BRCA2* or *BRCA1*, *BRCA2* and *ATM* (4-7%), that may not be identified by ctDNA test.^{4,5} We acknowledge that the FoundationOne® Liquid CDx test reports rearrangements and copy number variations for *BRCA1* and *BRCA2* but could miss a few *BRCA2* homozygous deletions.

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